



Technical Specifications of Power Conversion System

ENPCS-2520

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1. INTRODUCTION

Introducing our cutting-edge PCS, designed and manufactured in compliance with the industry-standard IEC/EN 62477-1, IEC/EN 61000-6-2 and IEC/EN 61000-6-4. These standards, developed by the International Electrotechnical Commission (IEC), ensure that our PCS is safe, reliable, and able to perform its function to the highest level of efficiency and quality.

IEC/EN 62477-1 is a standard for the safety of power electronic converter systems and equipment, which ensures that our PCS meets the safety requirements for power electronic systems and equipment.

IEC/EN 61000-6-2 covers the electromagnetic compatibility (EMC) requirements for residential, commercial, and light-industrial environments, meaning that our PCS is designed to operate with minimal interference to other electronic equipment and minimal emission of electromagnetic interference.

IEC/EN 61000-6-4 is for EMC requirements for industrial environments, which ensures that our PCS will operate effectively in harsh industrial environments, while also minimizing the emission of electromagnetic interference.

By adhering to these standards, we ensure that our PCS is safe, reliable, and able to perform its function to the highest level of efficiency and quality, while also being compliant with the latest international standards and regulations. This guarantees the performance and longevity of our product and guarantees that it will exceed your expectations.

Envision's energy storage PCS is comprised of a bidirectional DC / AC converter, a control unit and more. The PCS controller receives control instructions from the background through communication, directing the converter to charge or discharge batteries based on the power instructions' symbol and magnitude. This adjustment helps regulate the active and reactive power supplied to the grid. The PCS communicates with the BMS via a CAN interface and dry contact to obtain information about the bank's state. This facilitates protective charging and discharging of batteries, ensuring their safe operation.

1.1. PCS Functions

The PCS performs several functions, as outlined below:

- **On-Grid Operation:** During on-grid operation, the PCS autonomously monitors grid voltage and frequency, efficiently managing bidirectional converter control. It enables precise adjustment of both active and reactive power individually, allowing intelligent management of bank-level charging and discharging processes.
- **Protection:** The PCS incorporates various protection functions to ensure the safe, reliable, and continuous operation of the system. Key protection functions include short-circuit protection, DC reverse connection protection, battery overcurrent protection, hardware fault protection, and overload protection, among others.

- **Display and Communication:** A web monitoring interface is provided for parameter configuration and displaying system status and fault information. The equipment supports communication protocols such as Modbus RTU, Modbus TCP / IP, ensuring reliable communication with SCADA / EMS and the reception of power dispatch commands.
- **Other Functions:** The PCS incorporates functions such as power factor adjustment, reactive power compensation, and active power adjustment.

2. PCS PARAMETERS

Table 1: PCS parameters.

Item	ENPCS-2520
Parameters on AC System	
AC Connection	Three-phase three-wire
Rated Power	2520 kW
Overload Capacity	110%: 10 min@45 degree
	110%: Continuous @ 40 degree
	120%: 1 minute @ 35 degree
The Rated Voltage on the AC Side	690 V
Rated Current	2109 A
Rated Grid Frequency	50 Hz (consistent with grid frequency)
Current Waveform THD	Compliance with IEEE 519
Power Factor Range [λ]	-0.95-0.95
Power Control Deviation	$\leq 2\%$
DC Component	0.5% (at rated current)
Parameters on DC System	
DC Voltage Range	1000 – 1500 V
Full-Power DC Voltage Range	1165 – 1500 V
Voltage Regulation Accuracy	$\pm 2\%$
Current Regulation Accuracy	$\pm 2\%$
Protection	

LVRT	Yes
HVRT	Yes
Anti-Islanding Protection	Yes
AC Over Current / Short Circuit Protection	Yes
AC Over-Voltage / Undervoltage Protection	Yes
AC Over / Under-Frequency Protection	Yes
Phase Sequence Error Protection of AC Incoming Line	Yes
DC Over Current / Short Circuit Protection	Yes
DC Over Voltage / Undervoltage Protection	Yes
DC Reverse Polarity Protection	Yes
Over Temperature Protection	Yes
Power Module (IGBT) Protection	Yes
Communication Fault Protection	Yes
Cooling System Fault Protection	Yes
DC Component Over- Standard	Yes

Protection		
Fault Recording	Yes	
	No less than 4 cycles, 8 channels.	
System		
Power Response Speed	< 30 ms	From when a power dispatch instruction is received in the hot standby state.
Charge-Discharge Switchover Time	< 30 ms	Mutual switchover time from 100%. Charging at rated power to 100% discharging at rated power.
Enclosure Protection Class	IP65	
Cooling Method	Liquid cooling	
Communications Interface	Ethernet / CAN / RS485 (Standard: Modbus, TCP)	
Operating Environment		
Operating Temperature	-30 – 60°C (derating above 45°C)	
Storage Temperature	-40 – 70°C	
Allowable Relative Humidity	0 – 100%	
Altitude	No derating at ≤ 2000 m	
Compliance	IEC 62477, IEC61000, IEC 62909-1, IEC 62509, IEC 62109, IEC62271-202, EN 50549, IEC 50530, IEEE 519, G99 etc.	

3. INVERTER TECHNICAL CHARACTERISTICS

3.1. General Control Functionality

- The PCS is capable of receiving control instructions from the monitoring system to charge and discharge the battery.

- The PCS can process various alarms of the battery management system (BMS) to ensure battery safety. Charging and discharging will cease immediately if any warning signal is received from the BMS or if the terminal voltage of the battery unit reaches the maximum charging threshold. The PCS also automatically halts charging if the cell voltage monitoring circuit in the BMS fails or if communication between the PCS and the BMS is disrupted.
- Users have the option to schedule charging and discharging of the PCS using a timer.
- Both AC and DC sides of the PCS are equipped with breaker components and have emergency shutdown functions. The AC output side of each PCS is equipped with a circuit breaker to ensure safe isolation from the low-voltage side of the boost transformer.

The PCS has a complete set of protection features, including:

- Low voltage ride through.
- High voltage ride through.
- Island effect protection.
- AC overcurrent / short circuit protection.
- AC overvoltage / undervoltage protection.
- AC over frequency / underfrequency protection.
- AC input phase sequence error protection.
- DC overcurrent / short circuit protection.
- DC overvoltage / undervoltage protection.
- DC polarity reverse protection.
- Over-temperature protection.
- Power module (IGBT) protection.
- Communication fault protection.
- Cooling system fault protection.
- DC component out-of-specification protection.

The PCS device has the following features to support the power grid:

- The PCS can automatically synchronise with the power grid.

- The AC component uses LC filtering, in combination with the current repetition control technology to reduce output current harmonics, proportional-resonant control technology to suppress specific sub-grid-connected current harmonics, dead-zone compensation technology to reduce grid-connected current harmonics caused by IGBT switching dead-zone, and converter modulation algorithm switching technology to reduce grid-connected current harmonics under low load rate conditions to ensure that the quality of power delivered to the power grid meets the current power grid requirements to the maximum extent.
- Active power control.
 - When the active power command is a fixed value, the PCS outputs a constant power value. The active power does not vary with frequency and voltage under normal operating conditions.
 - The PCS can control its active power output by following the instructions from the energy storage energy management system. In standby mode, the time from receiving the power scheduling command to responding to the power output shall not exceed 100 ms, and the power output shall not exceed the maximum allowable power of the PCS.
- Voltage/reactive power regulation.
 - When the reactive power is constant, the PCS can output a constant power, and the reactive power does not vary with frequency and voltage.
 - The PCS can track and regulate the reactive power output in real time by following the control instructions from the monitoring system in the energy storage power station.
 - The power factor at rated power is adjustable between -1~1
- Necessary current limiting measures shall be taken at any stage of charging according to the needs of the battery to avoid damage.
- The PCS has high and low voltage ride-through capabilities.
- The PCS has a certain ability to tolerate system frequency anomalies.
- The PCS supports primary frequency modulation.

3.2. Power De-Rating due to Temperature

Temperature de-rating is the reduction of the inverter's maximum power output as temperature increases. This is necessary as the inverter performance typically decreases with temperature increases due to factors such as decreased component efficiency, increased conductor resistance, and reduced cooling capacity. The maximum operating temperature and a de-rating factor are used to calculate the maximum power output at a given temperature. This helps to prevent overheating and damage to the inverter and ensures stable power delivery to the grid.

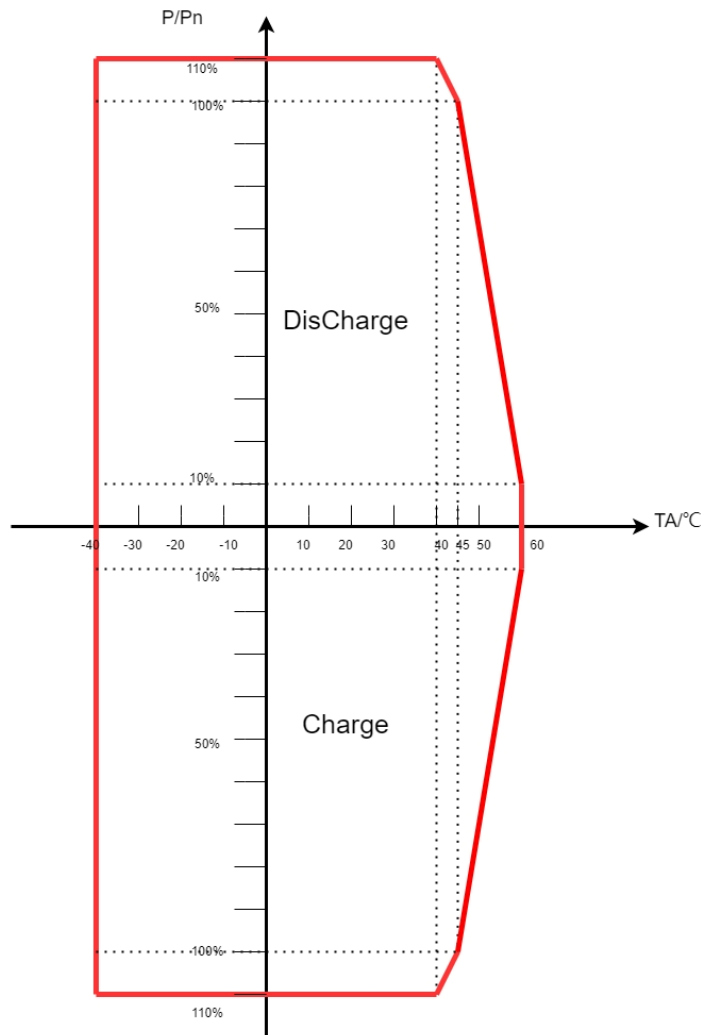


Figure 1: Temperature de-rating of Envision's inverter.

3.3. Power De-Rating due to Altitude

Inverter power de-rating due to altitude refers to the decrease in power output of the inverter as the altitude increases. This is because the air density decreases at higher altitudes, which reduces the cooling efficiency of the inverter. As a result, the inverter must be de-rated to prevent overheating and ensure safe operation.

Envision propose power de-rating when altitude is above 2000 m.

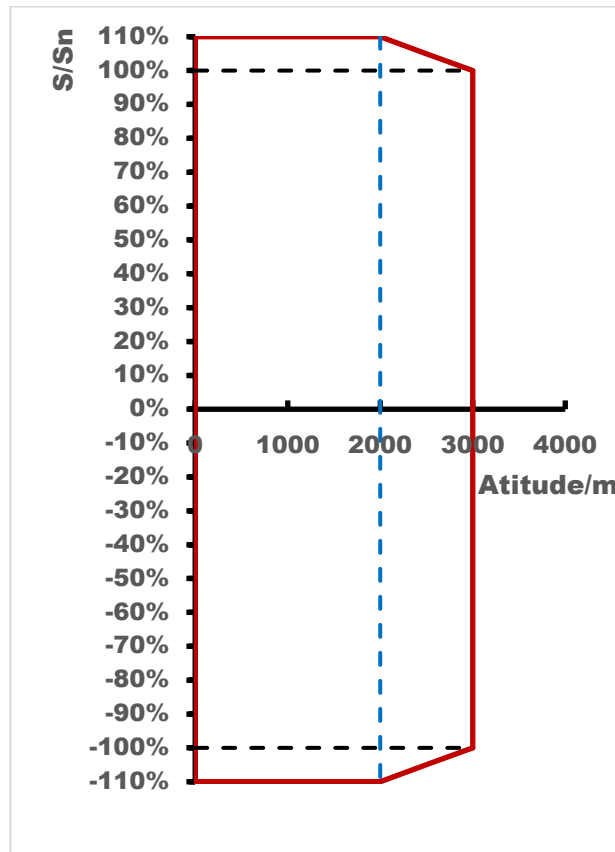


Figure 2: Altitude de-rating curve of Envision's inverter.

3.4. Power De-Rating due to Variations in the PCS Voltage

The inverter performance typically decreases with voltage increases due to factors such as decreased component efficiency, increased conductor resistance, and increased power losses. The maximum operating voltage and a derating factor are used to calculate the maximum power output at a given voltage. This helps to prevent damage to the inverter and improve power quality by reducing voltage fluctuations and harmonics. Note that for this current-source inverter (CSI), the voltage derating results in a higher power factor.

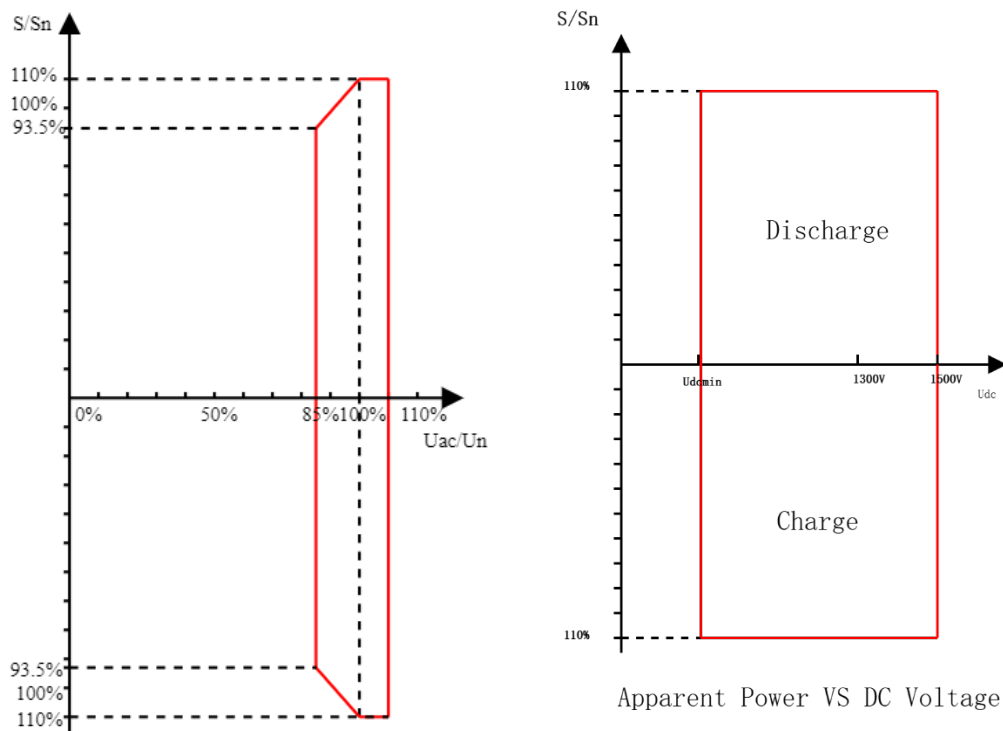


Figure 3: Apparent power vs AC voltage (left) and vs DC voltage (right).

3.5. Efficiencies

The PCS efficiency which determines how much net DC power is generated and is converted to AC power is generally variable. These variations depend on the loading, as well as the voltages of the inverter and rectifier. The rectifier and inverter efficiency as a function of varying DC voltage and operating power are given in Table 2.

Table 2: Inverter efficiency as a function of varying DC voltage

U _{dc} (V)	AC side power point	
	33% P _n	100% P _n
U _{max} (Discharge)	98.851%	98.396%
U _{mid} (Discharge)	98.965%	98.436%
U _{min} (Discharge)	99.053%	98.501%
U _{max} (Charge)	98.817%	98.244%
U _{mid} (Charge)	98.914%	98.297%
U _{min} (Charge)	99.004%	98.352%

3.6. PQ Curve

The PQ curve is a graphical representation of the inverter's operating range, showing the active power (P) and reactive power (Q) it can supply to the grid. It determines the inverter's operating limits and can be used to optimize its operation and assess its power quality.

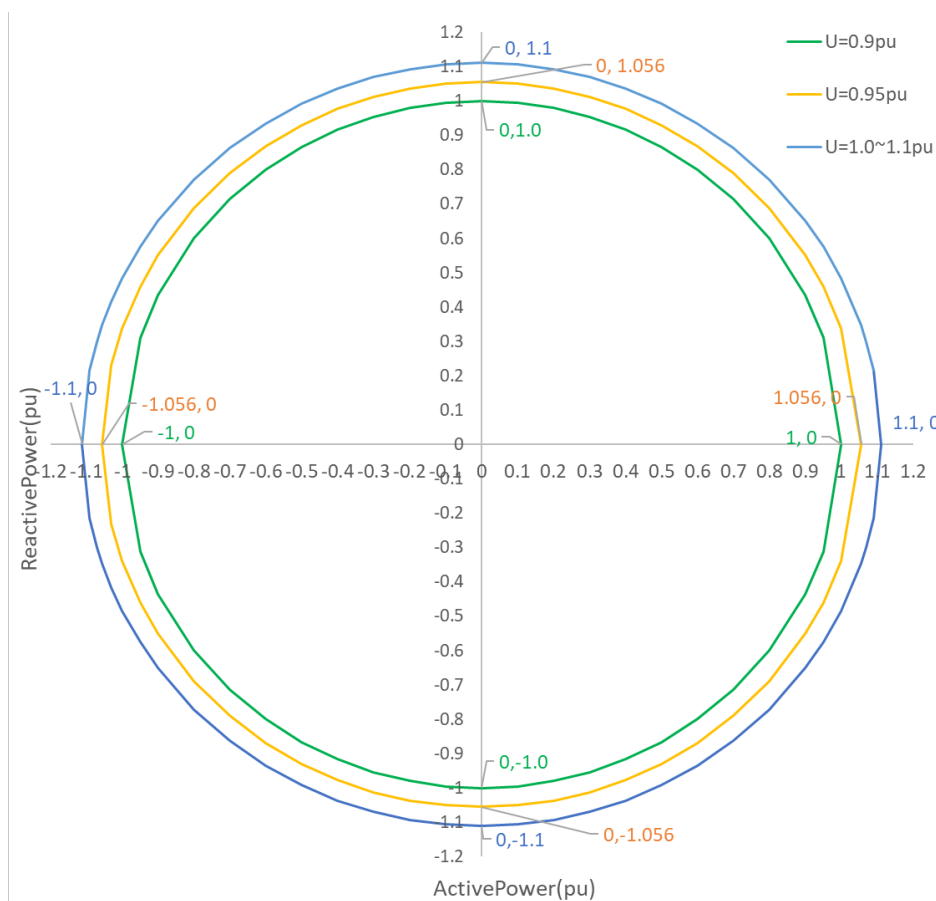


Figure 4: To avoid any doubt, the capability is limited by the apparent current which is 1.1 pu of rated current ($P=2520$ kW, $U=1$, Power factor=1). And the maximum apparent power is 110% of P rated (2520kW).

4. POWER QUALITY

4.1. Frequency Range

Table 3: Requirements for frequency ranges.

Frequency Range (Hz)	Requirements
< 47	The PCS shall not be charged. The PCS shall determine whether to separate from the grid based on the minimum allowable frequency or the requirements of the grid dispatch centre.
47 – 47.5	The PCS being charged shall be switched to discharging within 0.2 s. If it does not have the discharging condition or under other special circumstances, it shall be separated from the grid within 0.2 s. The PCS being discharged shall operate for 30 mins.
47.5 – 51.5	Operation with normal charging or discharging.
51.5 – 52	The PCS being discharged shall be switched to charging within 0.2 s. If it does not have the charging condition or under other special circumstances, it shall be separated from the grid within 0.2 s. The PCS being charged shall operate for 30 mins.
> 52	The PCS shall not be discharged. The PCS shall determine whether to separate from the grid based on the maximum allowable frequency.

NOTE: The PCS software parameters can be adjusted to the local grid code frequency protection requirements.

4.2. Voltage Range

Table 4: Runtime modes for voltage ranges.

Voltage Range (UN)	Runtime
110% – 130%	HVRT
90% – 110%	Continuous operation
0% – 90%	LVRT

NOTE: The PCS software parameters can be adjusted to the local grid code voltage protection requirements.

5. GRID FOLLOWING TECHNOLOGY AND CAPABILITY

5.1. Response Time and Control Accuracy

Response time and ramp rate are closely related but are distinct parameters that describe the performance of an energy storage system inverter. Response time, also known as reaction time, measures the amount of time it takes for an inverter to respond to a change in grid conditions. It is the time interval between a grid event (i.e. a disturbance, a drop or an increase in voltage) and the inverter's reaction (i.e. starting to increase or decrease power). Meanwhile, ramp rate measures the rate at which the power output of an inverter can change, usually measured in watts per second (W/s) or as a percentage of the rated power per second. The importance of both parameters is crucial for ensuring stability of the grid and protecting the equipment from damage. While both parameters are closely related, the main distinction between them lies in their focus: response time refers to the reaction time, while ramp rate is the change of power output.

The response time is defined in Figure 5.

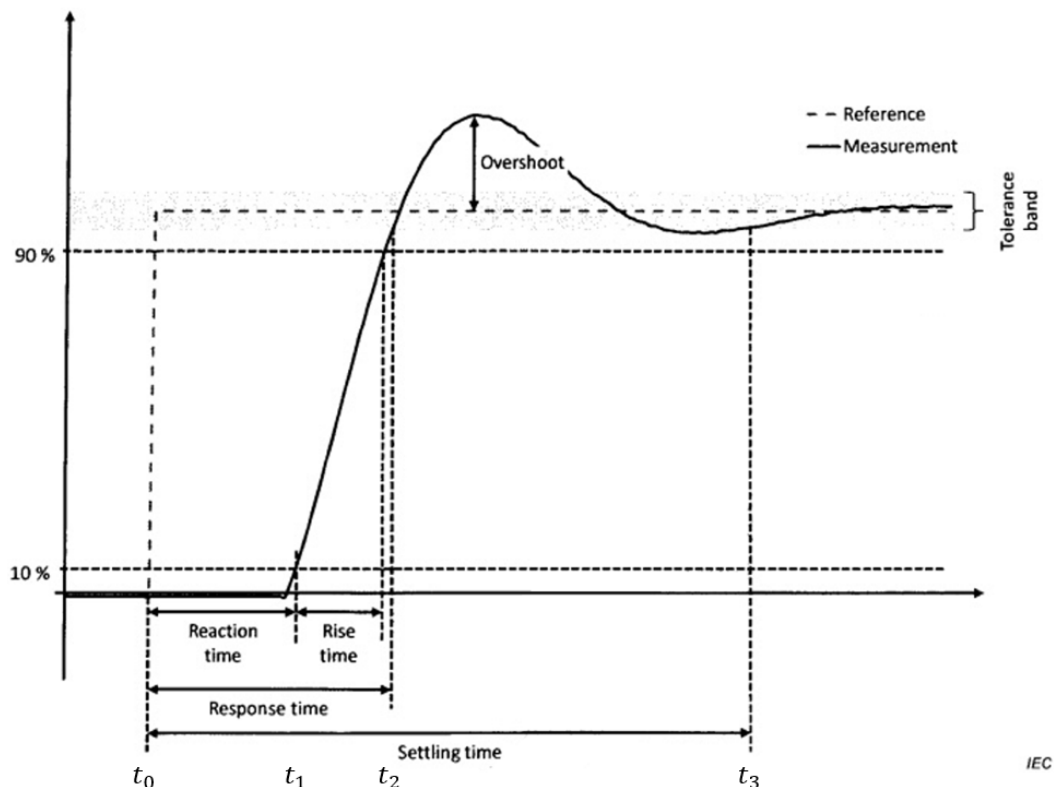


Figure 5: Example of step response.

- Reaction time is the time interval between when the power order is sent and the initial moment when the inverter starts to inject.
- Rise time is the time needed by the inverter to go from point t_1 (10% of the reference value) to point t_2 (90% of the reference value).
- Response time is the time interval between t_0 to t_2 .
- Settling time is the time between the moment in which the inverter reacts to power command (t_0) and the moment in which the power is stabilized within $\pm 2\%$ of the set point (t_3).

These times strongly depend on the inverter internal control configuration and the grid conditions (SCR and X/R ratio) measured at the inverter terminals.

Table 5: Dynamic response of Envisions PCS in grid following mode

Reaction Time	Response Time	Setting Time
≤ 5 ms	≤ 30 ms	≤ 50 ms

5.2. Harmonics

Power harmonics refer to the presence of unwanted frequencies in the electrical system that can cause problems such as equipment malfunction and power loss. Inverters, as power electronic devices, can generate harmonic currents that can affect the quality of power. Harmonics occur as integer multiples of the fundamental frequency which is typically 50 Hz or 60 Hz in electronic power grids. Harmonic currents cause voltage drops, which superimpose the nominal grid voltage resulting in distortion of the sine wave of the grid voltage. Harmonics can be generated by non-linear loads or from power electronic means with high frequent switching transistors (inverter for example). The International Electrotechnical Commission (IEC / EN) has established standards for power harmonics in IEEE 519. These standards include limits on the harmonic distortion levels, specifically, the Total Harmonic Distortion (THD) should not exceed 3% and the individual harmonic components should not exceed the limits specified in the standard.

Envision's PCS adheres with the IEEE 519 requirements concerning harmonics. For a comprehensive understanding of harmonic current injection, we can provide a detailed report upon request.

5.3. Voltage Control

It could work as a reactive power source and be controlled by the whole plant control system to control the voltage at the plant connection point.

5.4. Fast Frequency Control

PCS supports the primary frequency control (frequency droop control) and the inertia control. It could also work as an active power source and be controlled by the whole plant control system to release the frequency droop control at the plant connection point.

5.5. Islanding Detection “Anti-Islanding”

Islanding refers to the operation of a power grid as an isolated system, disconnected from external transmission lines. This can occur intentionally, as a planned outage or maintenance procedure, or unintentionally, due to a fault in the transmission system or a natural disaster. When a power grid is islanded, it must rely on local generation sources, such as solar panels or wind turbines, to meet the energy demands of the area. Islanding can be a challenge for power systems because it can be difficult to maintain a stable frequency and voltage without the support of external transmission lines. Envision’s inverters play a key role in islanded power systems by converting the direct current (DC) power produced by local generation sources into alternating current (AC) power, which can be used by consumers.

With the function of active islanding detection, the PCS could disconnect from the grid within 2 seconds after grid failure.

Figure 6 shows the field test data of active islanding detection. It shows the PCS disconnects from the grid within 840 ms.



Figure 6: Field test data of active islanding detection.

5.6. Low / High Voltage Ride-Through (LVRT / HVRT)

Voltage ride-through refers to the ability of the inverter to maintain its connection to the grid during voltage sags or dips. LVRT is the ability of the inverter to continue operating and supplying power to the grid during low voltage events, while HVRT is the ability to do the same during high voltage events. Maintaining a connection to the grid during voltage fluctuations is important for the stability and reliability of the power system.

The PCS can remain electrically connected to the utility grid system during certain voltage dip events (fault events). This functionality is known as 'Fault Ride Through'.

5.6.1. Low Voltage Ride Through (LVRT)

The profile that the PCS can 'ride through' remains within the limits shown in Figure 7.

If the voltage dip is deeper or lasts longer than the specified settings the PCS may be disconnected from the grid. Envision's PCS LVRT capability could cover global different requirements as shown in the figure below.

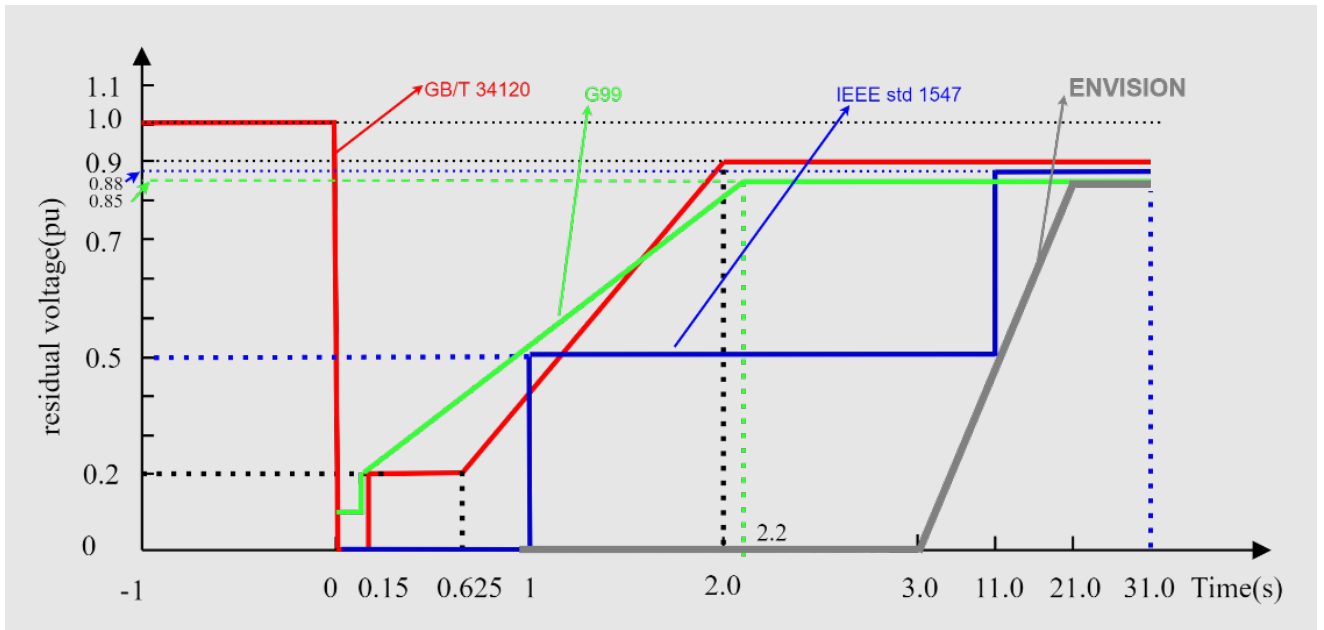


Figure 7: LVRT curve demonstrating that the PCS can 'ride through' low voltage events.

5.6.2. High Voltage Ride Through (HVRT)

The profile that the PCS can 'Ride Through' remains within the limits shown in Figure 8.

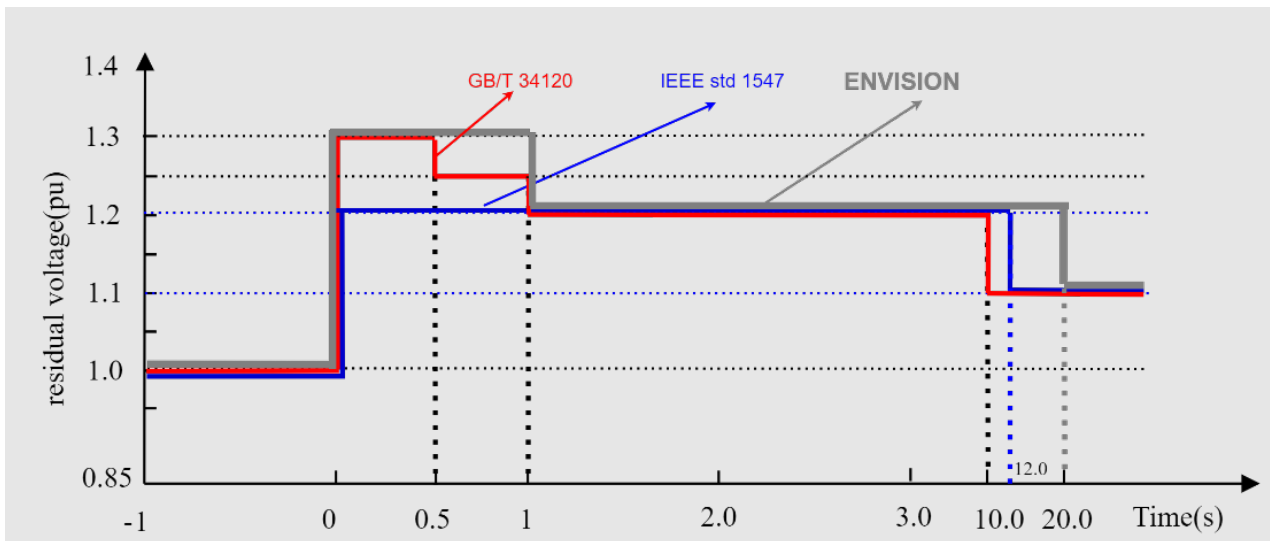


Figure 8: HVRT curve demonstrating that the PCS can 'ride through' high voltage events.

5.7. Short Circuit Contributions

When the grid experiences a failure, the AC voltage of the PCS decreases. To maintain stability, the inverter will enter an LVRT phase to supply reactive current to the power grid and control overcurrent. According to testing, when the inverter is in charging or discharging mode and the AC voltage drops below 10%, the maximum current that can be supplied by the inverter and injected into the grid is 1.2 times the rated current. To account for any margin of error in actual situations, it is considered that the maximum current supplied by the PCS during a grid failure is 1.3 times the rated current.

6. GRID FORMING TECHNOLOGY AND CAPABILITY

The PCS can run in both grid following and grid forming mode with seamless transition. It works as a virtual synchronous generator like a voltage source and supports active power control, reactive power control, virtual inertia, primary frequency control and voltage droop control. Also, the PCS has a current control in its inner loop to make sure the equipment operates safely.

The different application scenario for grid forming and grid following is displayed in the table below, while grid forming has irreplaceable advantages on weak grid and off grid modes.

Table 6: Application scenarios for grid forming and grid following.

	Grid Following	Grid Forming
Strong Grid	Applicable.	Applicable.
Weak Grid	Applicable. Limited transient response capability.	Applicable. Effective transient response.
Off Grid	Not Applicable.	Applicable.

The key technical functions of PCS for grid forming are shown below.

Table 7: Key functions of PCS for grid forming.

Functionalities	Corresponding Control Blocks
Frequency Support & Stabilisation	Frequency Governor Virtual Inertia
Voltage Support	Automatic Voltage Regulator
Current limitation during faults & inrush	Unit Control
Grid Strengthen when SCR < 2	Synthetic Impedance

Islanding Control

Unit Control

The LVRT simulation comparison results between grid forming (GFM) and the enhanced grid following (GFL) are shown below. The enhanced grid following is also referred to as grid supporting in some literature, which integrates the primary frequency and voltage regulation in the inverter to provide better frequency and voltage regulation during the transition. The comparison shows that grid frequency and voltage when grid forming works are much better than when the enhanced grid following is used.

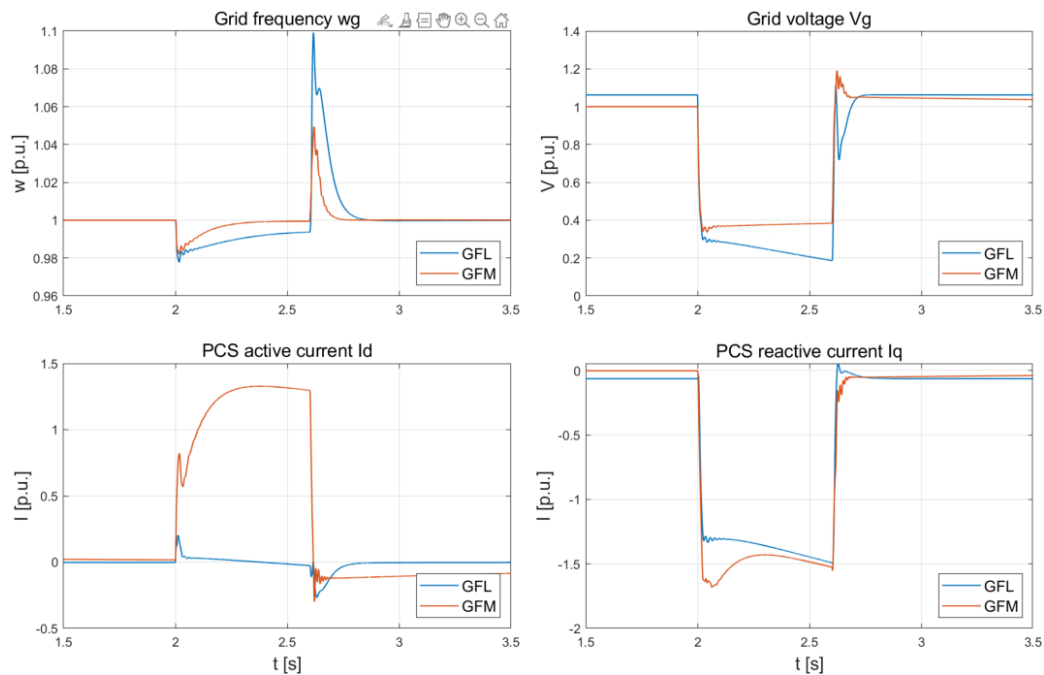


Figure 9: LVRT simulation results comparing grid forming and grid following.

Phase jump is an important grid disturbance scenario, especially in a renewable dominated power plant. The simulation result of 60° phase jump in a weak grid scenario for both GFM and enhanced GFL strategies are displayed in figure 10. The figure depicts grid frequency being stabilized by the virtual inertia of GFM.

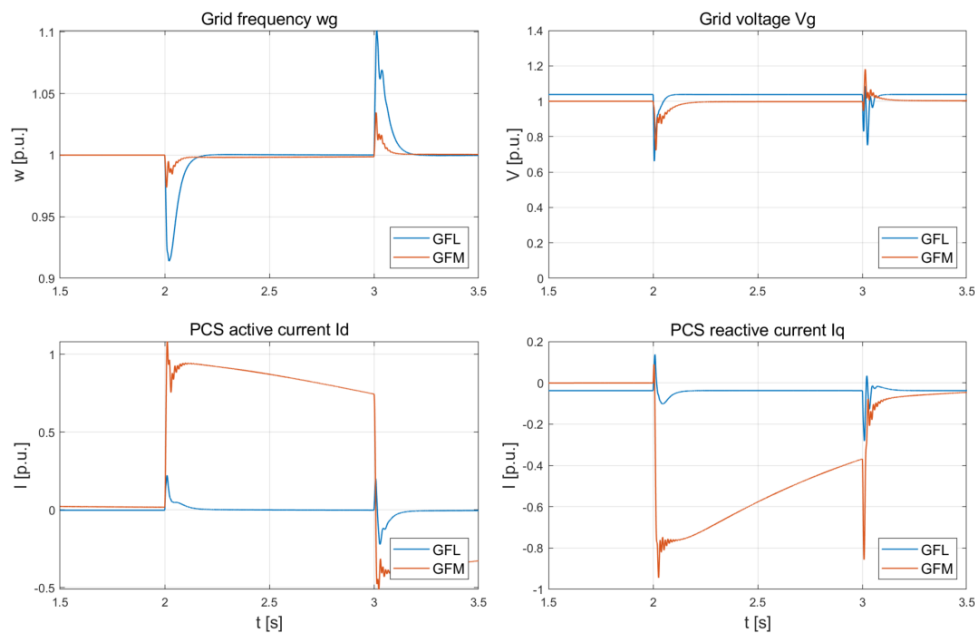


Figure10: 60° phase jump simulation results.

The simulation result of P load step change comparison between GFM and enhanced GFL is displayed in 11; GFM transient performance is seen to be much better than enhanced GFL.

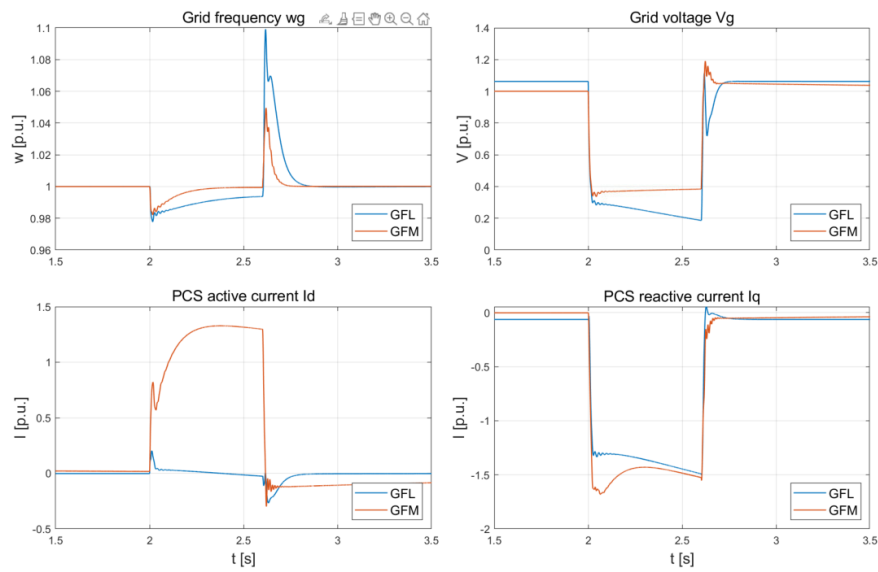


Figure 11: P load step change simulation results.

PCS can transit between GFM and enhanced GFL mode seamlessly. Figure 12 shows the seamless transition operation data. The PCS could keep the voltage, and the current of load has little disruption.

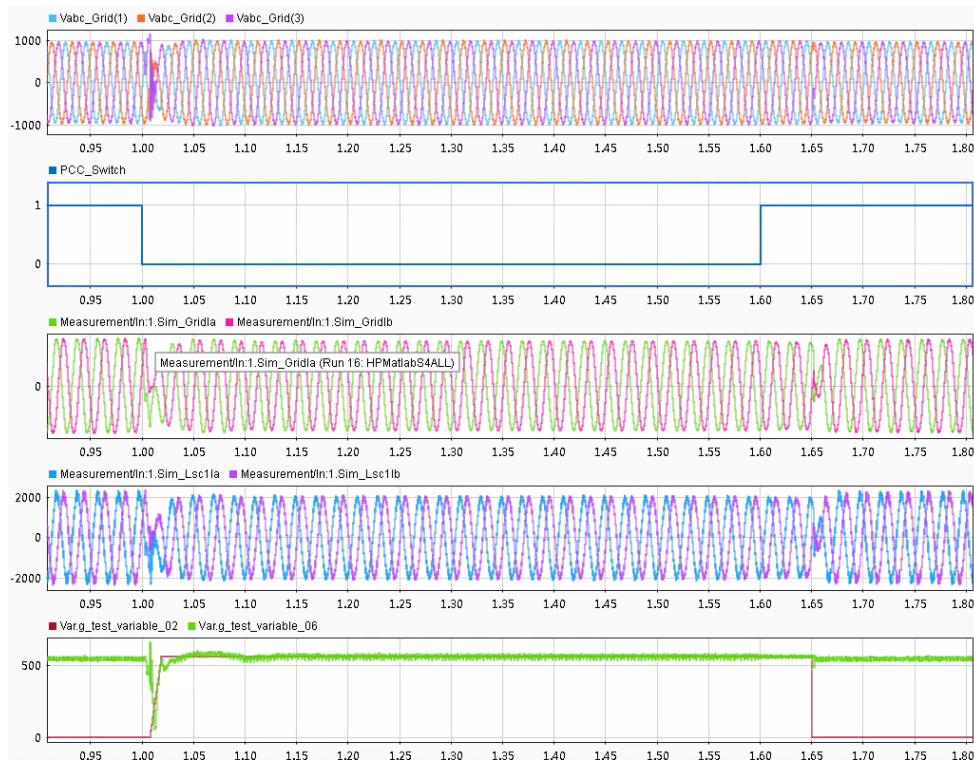


Figure12: Seamless transition of grid following and grid forming mode.

In discussions on grid-forming inverters, it's essential to note that most of them are implemented based on the power synchronization mechanism of synchronous machines. Consequently, classical stability issues such as power angle stability persist. Notably, the aggregation scale of the PCS is much larger than that of traditional synchronous power stations, while the average rated power of each PCS is much smaller. This scenario poses challenges for the stable operation of a large-scale aggregation of PCSs operating in VSG (Virtual Synchronous Generator) mode.

Envision employs virtual impedance emulation on PCSs based on large synchronous power station models, tailored to match weak grids or island-operating renewable energy systems. By detecting grid impedance, the output impedance of PCSs is adjusted to facilitate stabilized operation. Moreover, stability is further enhanced with the deployment of more PCSs based on the same impedance matching technology.

Envision has amassed considerable expertise in oscillation analysis and suppression, leveraging widely used impedance theory in China's renewable power sector. Figure 13 illustrates the simplified impedance theory. In scenarios with a high penetration rate of renewable energy in the grid, sources like wind and solar power, along with PCSs, may oscillate with the grid, particularly in weak grid conditions or grids with series-inductor compensators. Figure 14 demonstrates wind turbine oscillation current, which can be detrimental to both the grid and wind power systems.

Figure 15 depicts oscillation suppression achieved through matched impedance based on algorithm optimization.

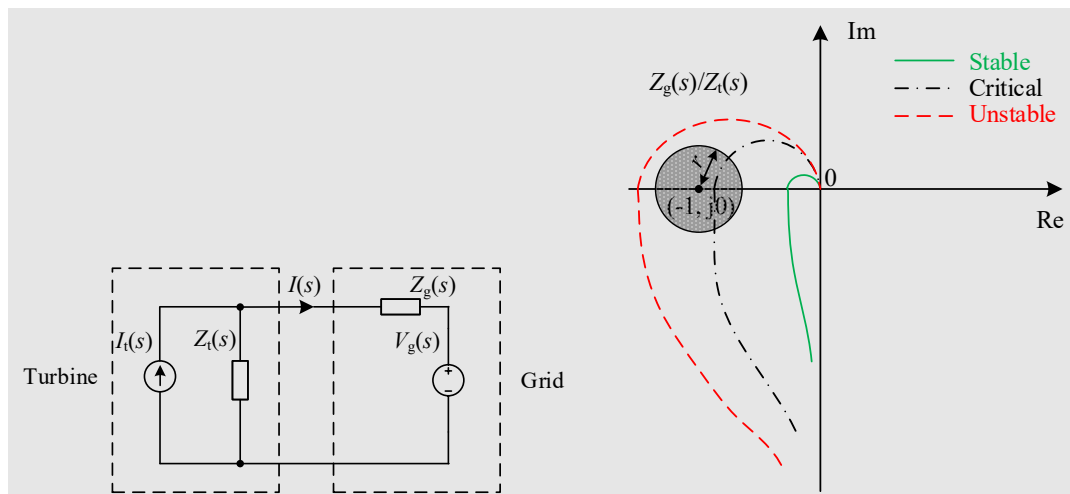


Figure13: System stability analysis based on impedance theory.

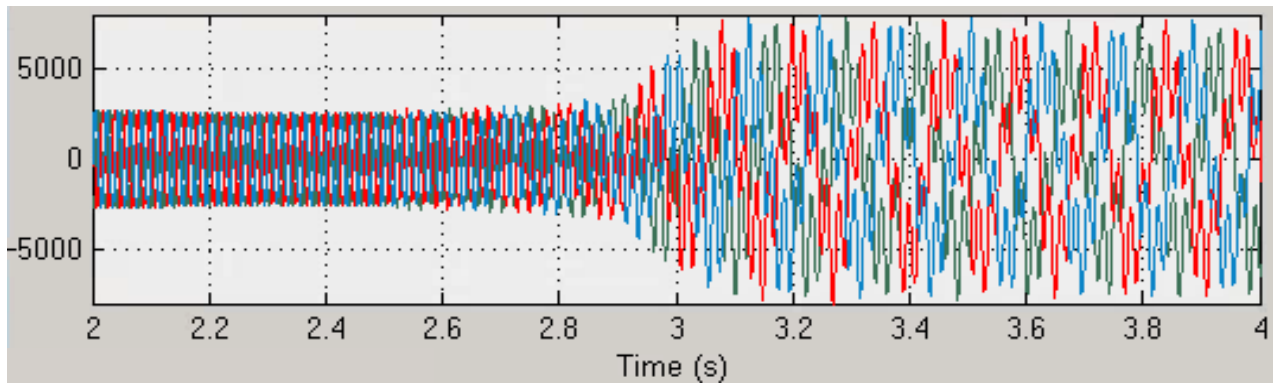


Figure 14 Wind turbine oscillation with grid.

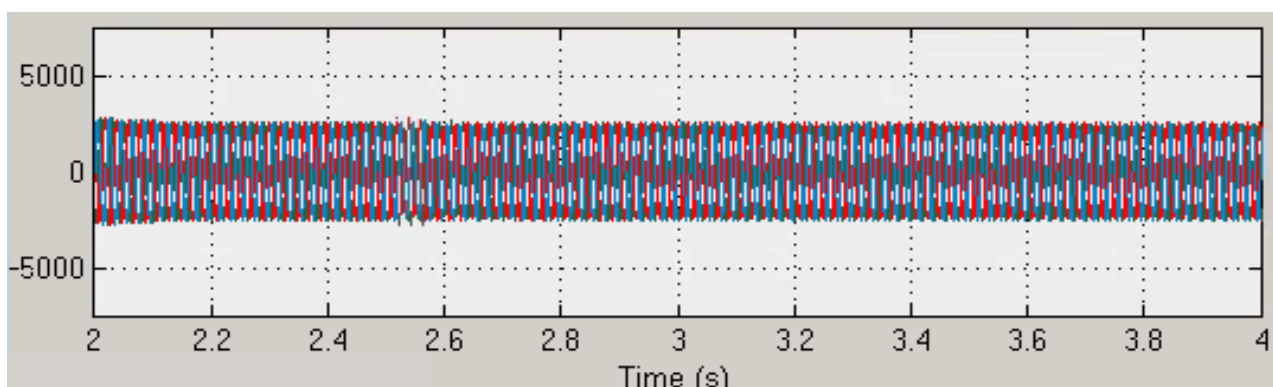


Figure 15: Oscillation suppression with matched impedance.